OPTIMIZATION - QUESTION POOL WITH ANSWERS

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1 Linear programming

1.1 Question

Represent the following LP problems in vector and matrix form. Which LP problem has a finite solution, no solution, or unbounded solution? Using graphical method and check the answer with CVXPY.

a.

$$\begin{aligned} \text{min. } & x_1 + x_2 \\ \text{s.t. } & x_1 - 2x_2 \geq 2, \\ & x_1 \geq 0, \\ & x_2 \geq 0. \end{aligned}$$

b.

$$\begin{aligned} \text{min. } & x_1 - x_2 \\ \text{s.t. } & x_1 + 2x_2 \leq -1, \\ & x_2 \geq 3. \end{aligned}$$

c.

$$\begin{aligned} \text{max. } & x_1 - x_2 \\ \text{s.t. } & 3x_1 + x_2 \leq 3, \\ & x_1 + x_2 \geq 1 \\ & x_2 \geq 3. \end{aligned}$$

1.2 Complementary slackness

Given the LP:

$$\begin{aligned} \text{max. } & -x_1 - x_2 \\ \text{s.t. } & -2x_1 + 7x_2 \leq 6 \\ & -3x_1 + 1x_2 \leq -1 \\ & 9x_1 - 4x_2 \leq 6 \\ & x_1 - x_2 \leq 1 \\ & 7x_1 - 3x_2 \leq 6 \\ & -5x_1 + 2x_2 \leq -3 \\ & x_1, x_2 \geq 0. \end{aligned}$$

a) Solve the primal problem using cvxpy, find the optimal dual variables?

b) Using complementary slackness theorem to show the relationship between the primal's constraints and the dual variables?

1.3 Vanderbei's book

A steel company must decide how to allocate next week's time on a rolling mill, which is a machine that takes unfinished slabs of steel as input and can produce either of two semi-finished products: bands and coils. The mill's two products come off the rolling line at different rates:

- Bands: 200 tons/hr
- Coils: 140 tons/hr

They also produce different profits:

- Bands: \$25/ton
- Coils: \$30/ton.

Based on currently booked orders, the following upper bounds are placed on the amount of each product to produce:

- Bands: 6000 tons
- Coils: 4000 tons.

Given that there are 40 hours of production time available this week, the problem is to decide how many tons of bands and how many tons of coils should be produced to yield the greatest profit. Formulate this problem as a linear programming problem.

1.4 Vanderbei's book

A small airline, Ivy Air, flies between three cities: Ithaca, Newark, and Boston. They offer several flights but, for this problem, let us focus on the Friday afternoon flight that departs from Ithaca, stops in Newark, and continues to Boston. There are three types of passengers:

a) Those travelling from Ithaca to Newark.

b) Those travelling from Newark to Boston.

c) Those travelling from Ithaca to Boston.

The aircraft is a small commuter plane that seats 30 passengers. The airline offers three fare classes:

a) Y class: full coach.

b) B class: nonrefundable.

c) M class: nonrefundable, 3-week advanced purchase.

Ticket prices, which are largely determined by external influences (i.e., competitors), have been set and advertised as follows:

	Ithaca–Newark	Newark–Boston	Ithaca–Boston
Y	\$300	\$160	\$360
B	\$220	\$130	\$280
M	\$100	\$80	\$140

Based on past experience, demand forecasters at Ivy Air have determined the following upper bounds on the number of potential customers in each of the 9 possible origin-destination/fare-class combinations:

	Ithaca–Newark	Newark–Boston	Ithaca–Boston
Y	4	8	3
B	8	13	10
M	22	20	18

The goal is to decide how many tickets from each of the 9 origin/destination/fare-class combinations to sell. The constraints are that the plane cannot be overbooked on either of the two legs of the flight and that the number of tickets made available cannot exceed the forecasted maximum demand. The objective is to maximize the revenue. Formulate this problem as a linear programming problem.

Proof. Let's:

- x_{11}

Objective:

$$300 * x_{11} + 160 * x_{12} + 360 * x_{13} + 220 * x_{21} + \ldots + 80 * x_{32} + 140 * x_{33}$$

Constraints:

$$xij \geq 0; x11 \leq 4; x12 \leq 8; x13 \leq 3; \ldots; x33 \leq 18$$

□

1.5 Problem 4.1, [Boyd et.al.]

Consider the optimization problem

$$\text{minimize } f_0(x_1,\ x_2) \tag{1}$$
$$\text{subject to } 2x_1 + x_2 \geq 1 \tag{2}$$
$$x_1 + 3x_2 \geq 1 \qquad x_1 \geq 0,\ x_2 \geq 0 \tag{3}$$

Make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value.

a) $f_0(x_1,\ x_2) = x_1 + x_2$.

b) $f_0(x_1,\ x_2) = -x_1 - x_2$.

c) $f_0(x_1,\ x_2) =\ x_1$.

Proof. The feasible set is the convex hull

a) $x^* = (2/5, 1/5)$

b) Unbounded below

c) $X_{opt} = \{(0, x_2) \| x_2 \geq 1\}$.

□

1.6 Problem 4.8-d, Boyd et.al.

Minimizing a linear function.

$$\text{minimize } c^T x \tag{4}$$
$$\text{subject to } 1^T x = 1, \tag{5}$$
$$x \geq 0. \tag{6}$$

What happens if the equality constraint is replaced by an inequality $1^T x \leq 1$?

Proof. Suppose the components of c are sorted in increasing order with $c_1 = c_2 = \ldots = c_k < c_{k+1} \leq \ldots \leq c_n$: We have

$$c^T x \geq c_1(1^T x) = c_{\min}$$

for all feasible x, with equality if and only if $x_1 + \ldots + x_k = 1$, $x_1 \geq 0, \ldots, x_k \geq 0$, $x_{k+1} = \ldots = x_n = 0$. We conclude that the optimal value is $p* = c_1 = c_{\min}$. In the investment interpretation this choice is quite obvious. If the returns are fixed and known, we invest our total budget in the investment with the highest return.

If we replace the equality with an inequality, the optimal value is equal to

$$p^* = \min\{0, c_{\min}\}$$

(If $c_{\min} \leq 0$, we make the same choice for x as above. Otherwise, we choose $x = 0$.) □

1.7 Problem 4.8-e, Boyd et.al.

Minimizing a linear function.

$$\text{minimize } c^T x \tag{7}$$
$$\text{subject to } 1^T x = \alpha, \tag{8}$$
$$0 \leq x \leq 1. \tag{9}$$

where α is an integer between 0 and n. What happens if α is not an integer (but satisfies $0 \leq \alpha \leq n$)? What if we change the equality to an inequality $1^T x \leq \alpha$.

1.8 Problem 4.8-f, Boyd *et.al.*, fractional Knapsack problem

Minimizing a linear function over a unit box with a weighted budget constraint. (fractional Knapsack problem)

$$\text{minimize } c^T x \tag{10}$$

$$\text{subject to } d^T x = \alpha,\ 0 \leq x \leq 1 \tag{11}$$

with $d > 0$ and $0 \leq \alpha \leq 1^T d$.

Proof. We make a change of variables $y_i = d_i x_i$, and consider the problem

$$\text{minimize } P_{i=1}^n (c_i/d_i) y_i \tag{12}$$

$$\text{subject to } 1^T x = \alpha, 0 \leq y \leq d. \tag{13}$$

Suppose the ratios $c_i = d_i$ have been sorted in increasing order:

$$\frac{c_1}{d_1} \leq \frac{c_2}{d_2} \leq \ldots \leq \frac{c_n}{d_n}$$

To minimize the objective, we choose

$$y_1 = d_1, y_2 = d_2, ..., y_k = d_k$$

$$y_{k+1} = \alpha - (d_1 + \ldots + d_k), y_{k+2} = \ldots = y_n = 0$$

where $k = \max\{i \in \{1, 2, \ldots, n\} \| d_1 + \ldots + d_i \leq \alpha$ (and $k = 0$ if $d_1 > \alpha$). In terms of the original variables, $x_1 = \ldots = x_k = 1, x_{k+1} = (\alpha - (d_1 + \ldots + d_k)) = d_{k+1}, x_{k+2} = \ldots = x_n = 0$. □

4.12 Boyd et.al., Network flow problem

Consider a network of n nodes, with directed links connecting each pair of nodes. The variables in the problem are the flows on each link: x_{ij} will denote the flow from node i to node j. The cost of the flow along the link from node i to node j is given by $c_{ij} x_{ij}$, where c_{ij} are given constants. The total cost across the network is

$$C = \sum_{i,j=1}^{n} c_{ij} x_{ij}$$

Each link flow x_{ij} is also subject to a given lower bound l_{ij} (usually assumed to be non-negative) and an upper bound u_{ij}. The external supply at node i is given by b_i, where $b_i > 0$ means an external flow enters the network at node i, and $b_i < 0$ means that at node i, an amount b_i flows out of the network. We assume that $1^T b = 0$, i.e., the total external supply equals total external demand. At each node we have conservation of flow: the total flow into node i along links and the external supply, minus the total flow out along the links, equals zero. The problem is to minimize the total cost of flow through the network, subject to the constraints described above. Formulate this problem as an LP.

Answer

This can be formulated as the LP

$$\text{minimize } C = \sum_{i,j=1}^{n} c_{ij} x_{ij} \tag{14}$$

$$\text{subject to } b_i + \sum_{j=1}^{n} x_{ij} - \sum_{j=1}^{n} x_{ji} i = 1, \ldots, n \tag{15}$$

$$l_{ij} \leq x_{ij} \leq u_{ij}. \tag{16}$$

1.9 LP formulation

A company want to produce three kinds of products A, B, and C.

- Each unit of Product A is of weight 1, size 2, cost 4 and value 3.
- Each unit of Product B is of weight 1, size 2, cost 1 and value 2.
- Each unit of Product C is of weight 3, size 5, cost 2 and value 4.

Due to the limit resources, the total weight of products is limited by 30, the total size is limited by 24 and the total cost is limited by 36 units.

a) Write a linear program to maximize the total value of the products. Note that the products can be divided into fractional unit.

b) Write the LP problem in the matrix form. Solve the problem. Hint. You can use solver like CVXPY.

c) Write the dual problem for the problem in Problem 1?

d) write the complementary slackness. Which constraint can be relaxed to increase the total value?

1.10 LP - Alloy

A manufacturer wishes to produce an alloy that is, by weight, 30% metal A and 70% metal B. Five alloys are available at various prices as indicated below:

Alloy	1	2	3	4	5
%A	10	25	50	75	95
%B	90	75	50	25	5
Price/lb ($)	5	4	3	2	1.5

The desired alloy will be produced by combining some of the other alloys. The manufacturer wishes to find the amounts of the various alloys needed and to determine the least expensive combination. Formulate this problem as a linear program

1.11 LP - Oil refinery

An oil refinery has two sources of crude oil: a light crude that costs \$35/barrel and a heavy crude that costs \$30/barrel. The refinery produces gasoline, heating oil, and jet fuel from crude in the amounts per barrel indicated in the following table:

	Gasoline	Heating oil	Jet fuel
Light crude	0.3	0.2	0.3
Heavy crude	0.3	0.4	0.2

The refinery has contracted to supply 900,000 barrels of gasoline, 800,000 barrels of heating oil, and 500,000 barrels of jet fuel. The refinery wishes to find the amounts of light and heavy crude to purchase so as to be able to meet its obligations at minimum cost. Formulate this problem as a linear program.

1.12 LP

A small firm specializes in making five types of spare automobile parts. Each part is first cast from iron in the casting shop and then sent to the finishing shop where holes are drilled, surfaces are turned, and edges are ground. The required worker-hours (per 100 units) for each of the parts of the two shops are shown below:

Part	1	2	3	4	5
Casting	2	1	3	3	1
Finishing	3	2	2	1	1

The profits from the parts are \$30, \$20, \$40, \$25, and \$10 (per 100 units), respectively. The capacities of the casting and finishing shops over the next month are 700 and 1000 worker-hours, respectively. Formulate the problem of determining the quantities of each spare part to be made during the month so as to maximize profit.

1.13 Duality of LP

[Problems 5.1, 5.5, Vanderbei] What is the dual of the following linear programming problems:

a)

$$\begin{aligned} \text{maximize } & x_1 - 2x_2 && (17)\\ \text{subject to } & x_1 + 2x_2 - x_3 + x_4 \geq 0 && (18)\\ & 4x_1 + 3x_2 + 4x_3 - 2x_4 \leq 3 && (19)\\ & -x_1 - x_2 + 2x_3 + x_4 = 1 && (20)\\ & x_2, x_3 \geq 0. && (21) \end{aligned}$$

b)

$$\text{maximize } 2x_1 + 8x_2 - x_3 - 2x_4 \tag{22}$$
$$\text{subject to } 2x_1 + 3x_2 + 6x_4 \leq 6 \tag{23}$$
$$-2x_1 + 4x_2 + 3x_3 \leq 1.5 \tag{24}$$
$$3x_1 + 2x_2 - 2x_3 - x_4 \leq 4 \tag{25}$$
$$x_1, x_2, x_3, x_4 \geq 0. \tag{26}$$

Analyze meaning of the optimal dual variables. Write the complementary slackness property.

1.14 ILP: Min-cut problem

Given graph $G = (V, E, C)$. A cut (S, T) of graph G is a partition such that source node $s \in S$ and sink node $t \in T$. The capacity of a cut is defined as

$$c(S, T) = \sum_{(i,j)\in E} b_{ij} c_{ij}$$

Where $b_{ij} = 1$ if node $i \in S$ and node $j \in T$, $y_{ij} = 0$, otherwise. Min-cut problem is to find a cut with minimum capacity:

$$\text{minimize } \sum_{(i,j)\in E} b_{ij} c_{ij} \tag{27}$$
$$\text{subject to } x_s = 1, x_t = 0 \tag{28}$$
$$b_{ij} \geq x_i - x_j \tag{29}$$
$$b_{ij},\ x_i, x_j \in \{0, 1\}, \forall (i, j) \in E \tag{30}$$

Explain what the variables x_i and x_j for all $(i, j) \in E$ mean and why the introduction of these variables is necessary?

The min-cut problem is an integer linear program (ILP), because its variables take integer values. Because ILPs are generally difficult to solve, they are often relaxed to LPs. Consider the following relaxation of the integer constraints:

$$\text{minimize}_{b\in R^{|E|}, x\in R^{|V|}} \sum_{(i,j)\in E} b_{ij} c_{ij} \tag{31}$$
$$\text{subject to } x_s - x_t \geq 1, b \geq 0 \tag{32}$$
$$b_{ij} \geq x_i - x_j, \forall (i, j) \in E \tag{33}$$

a) Explain why we can view this as a relaxation. How does the optimal value of the original ILP compare to the optimal value of the relaxed LP?

b) Next, derive the dual of (31). Use the following dual variables: $f, y \in R^{|E|}$, $w \in R$ corresponding to the constraints in the order they appear in (31).

c) What does each constraint of the dual you derived in part b) mean? Hint: the dual of the relaxed min-cut problem is called max-flow problem. Interestingly, a well-known theorem (the max-flow min-cut theorem) tells us is that the original ILP and the max flow problem have equal optimal criterion values. Please verify this with an particular graph.

2 Convex sets, convex functions

2.1 Question

Show that a polyhedron $x \in R^n : Ax \leq b$, for some $A \in R^{m \times n}$, $b \in R^m$, is convex and closed.

Answer Intersection of convex sets is convex.

2.2 A family of concave utility functions.

A family of concave utility functions. For $0 < \alpha \leq 1$, let

$$u_\alpha(x) = \frac{x^\alpha - 1}{\alpha}, \forall x \geq 0$$

We also define

$$u_0(x) = \log x, x > 0$$

.

a) Show that for $x > 0$, $u_0(x) = \lim_{\alpha \to 0} u_\alpha(x)$.

b) Show that u_α are concave, monotone increasing, and all satisfy $u_\alpha(1) = 0$. These functions are often used in economics to model the benefit or utility of some quantity of goods or money. Concavity of u_α means that the marginal utility (i.e., the increase in utility obtained for a fixed increase in the goods) decreases as the amount of goods increases. In other words, concavity models the effect of satisfaction.

c) Some researches use the function $u_\alpha(x) = \frac{x^{1-\alpha}}{(1-\alpha)}$ if $\alpha > 0, \alpha \neq 1$, and $u_\alpha(x) = \log(x)$, if $\alpha = 1$. This function has the same properties as the above problem. Plot the function with $\alpha = 0,\ 1,\ 2,\ 500$.

Answer

a) In this limit, both the numerator and denominator go to zero, so we use l'Hopital's rule:

$$\lim_{\alpha \to 0} u_\alpha 0 = \lim_{\alpha \to 0} \frac{(x^\alpha - 1)'}{\alpha'} = \lim_{\alpha \to 0} x^\alpha \log x = \log x.$$

b) First derivative $u'_\alpha(x) = x^{\alpha-1}$, which is positive for all x (since $0 < \alpha < 1$), so these functions are increasing.

Second derivative: $u''_\alpha(x) = (\alpha - 1)x^{\alpha-2}$. Since this is negative for all x, we conclude that u_α is strictly concave.

2.3 Entropy

Prove that the entropy function $f(x) = -\sum_i x_i \log(x_i)$, where $x_i \in R,\ x_i > 0,\ \sum_i x_i = 1$ is concave.

Answer $f_i(x_i) = x_i \log(x_i)$. $f_i'' = -\frac{1}{x_i} < 0$. Sum of concave fuctions is concave.

2.4 Entropy, Kullback-Leibler divergence and the information inequality [Boyd et al.]

Two vector $u, v \in R^n$. The relative entropy of two vectors is defined as

$$\sum_{i=1}^{n} u_i \log(u_i/v_i)$$

is convex on $(u, v) \in R^{2n}$.

Answer Consider the convex function $f(x) = -\log x, x \in R, x > 0$. Its perspective is

$$g(x,t) = -t\log(x/t) = t\log(t/x) = tlogt - t\log x,$$

and is convex for all $x > 0$ and $t > 0$. The function g is called the relative entropy of t and x. For $x = 1$, g reduces to the *negative entropy function.*

The relative entropy of u, v is convex since it is a sum of relative entropies of u_i, v_i.

2.5 Kullback-Leibler divergence

The Kullback-Leibler divergence D_{kl} defined as:

$$D_{kl}(u,v) = \sum_{i=1}^{n} (u_i \log(u_i/v_i) - u_i + v_i)$$

is convex.

Prove the information inequality: $D_{kl}(u, v) \geq 0$ for all $u, v \in R^n_{++}$. Also show that $D_{kl}(u, v) = 0$ if and only if $u = v$. It can be used as a measure of deviation between two positive vectors.

Answer $D_{kl}(u, v)$ is convex, since it is the relative entropy plus a linear function of (u, v).

The Kullback-Leibler divergence can be expressed as

$$D_{kl}(u,v) = f(u) - f(v) - \nabla f(v)^T (u - v),$$

where $f(v) = \sum_{i=1}^{n} v_i \log v_i$ is the negative entropy of v.

Since negative entropy is strictly convex,

$$f(u) > f(v) + \nabla f(v)^T (u - v)$$

. Hence, $D_{kl}(u, v) > 0$.

2.6 Convex, concave, quasi-convex, quasi-concave

For each of the following functions determine whether it is convex, concave, quasiconvex, or quasiconcave.

1. $f(x) = e^x - 1$
2. $f(x_1, x_2) = x_1 x_2$ on R^2_{++}
3. $f(x_1, x_2) = 1/x_1 x_2$ on R^2_{++}
4. $f(x_1, x_2) = x_1/x_2$ on R^2_{++}
5. $f(x_1, x_2) = x_1^2/x_2$ on $R \times R_{++}$
6. $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ on $R^2_{++}, 0 \leq \alpha \leq 1$

Answer

b) Hessian matrix:

$$\nabla^2 f(x) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

which is neither positive semidefinite nor negative semidefinite. Hence, it is neither convex nor concave.

It is quasi-concave.

c) Hessian matrix is positive definite

$$\nabla^2 f(x) = \frac{1}{x_1 x_2} \begin{bmatrix} \frac{2}{x_1^2} & \frac{1}{x_1 x_2} \\ \frac{1}{x_1 x_2} & \frac{2}{x_2^2} \end{bmatrix}.$$

Therefore, f is convex.

2.7 Monotonicity of the gradient

Let f be twice differentiable, with $dom(f)$ convex. Prove that f is convex if and only if $(\nabla \mathrm{f}(x) - \nabla \mathrm{f}(y))^T (x - y) \geq 0$, for all x, y. This property is called monotonicity of the gradient ∇f.

Answer From the convexity of f:

$$f(x) \geq f(y) + \nabla f(y)^T (x - y)$$

$$f(y) \geq f(x) + \nabla f(y)^T (y - x)$$

Add two inequalities will yield the answer.
The converse statement is not true.

2.8 Norm

Composition with an affine function. Show that the function $f(x) = ||Ax - b||$, where $A \in R^{m \times n}$, $b \in R^m$ and $||.||$ is a norm on R^m is convex.

Answer f is the composition of a norm, which is convex, and an affine function.

2.9 Pointwise maximum and supremum

Show that the following functions $f(x) = \max_{i=1,\ldots,k} ||A^{(i)}x - b^{(i)}||$ where $A^{(i)} \in R^{m \times n}$, $b^{(i)} \in R^m$ and $||.||$ is a norm on R^m is convex.

Answer f is the pointwise maximum of k functions $||A^{(i)}x - b^{(i)}||$. Each of those functions is convex because it is the composition of an affine transformation and a norm.

2.10 Log-concave functions

Show that the following functions are log-concave

a) Logistic function $f(x) = \frac{e^x}{1+e^x}$

b) Harmonic mean $f(x) = \frac{1}{\frac{1}{x_1}+\ldots+\frac{1}{x_n}}, x_i > 0, \in R$

3 Convex problems

3.1 [P4.1, Boyd et.al.]

Consider the optimization problem

$$\begin{aligned} &\text{minimize } f_0(x_1,\ x_2) \\ &\text{subject to } 2x_1 + x_2 \geq 1 \\ &\qquad\qquad x_1 + 3x_2 \geq 1 \\ &\qquad\qquad x_1 \geq 0,\ x_2 \geq 0 \end{aligned}$$

Make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value.

a) $f_0(x_1,\ x_2) = \max\{x_1, x_2\}$.

b) $f_0(x_1,\ x_2) = x_1^2 + x_2^2$.

Proof. a) $x^* = (1/3, 1/3)$

□

3.2 Problems involving ℓ_∞- and ℓ_1-norms.

Formulate the following problems as LPs. Explain in detail the relation between the optimal solution of each problem and the solution of its equivalent LP.

a) Minimize $\|Ax - b\|_\infty$ (ℓ_∞-norm approximation).

b) Minimize $\|Ax - b\|_1$ (ℓ_1-norm approximation).

c) Minimize $\|Ax - b\|_1$ subject to $\|x\|_\infty \leq 1$.

d) Minimize $\|x\|_1$ subject to $\|Ax - b\|_\infty \leq 1$.

e) Minimize $\|Ax - b\|_1 + \|x\|_\infty$.

In each problem, $A \in R^{m\times n}$ and $b \in R^m$ are given.

Proof. a) Equivalent to the LP

$$\begin{aligned} &\text{minimize } t \\ &\text{subject to } a_k^T x - b_k \leq t, k = 1, \ldots, m \\ &\qquad\qquad a_k^T x - b_k \geq -t, k = 1, \ldots, m. \end{aligned}$$

b) Equivalent to the LP

$$\text{minimize } s_1 + \ldots + s_m \text{subject to} \qquad a_k^T x - b_k \leq s_k, k = 1, \ldots, m$$
$$a_k^T x - b_k \geq -s_k, k = 1, \ldots, m.$$

The objective function of the LP is separable, so we achieve the optimum at

$$s_k = |a_k^T x - b_k|.$$

Therefore, optimizing over t and s simultaneously is equivalent to the original problem.

c) Equivalent to the LP

$$\begin{aligned} \text{minimize } & 1^T y = y_1 + \ldots + y_m \\ \text{subject to } & -y \leq Ax - b \leq y \\ & -1 \leq x \leq 1. \end{aligned}$$

with variables $x \in R^n$ and $y \in R^m$.

d) Equivalent to the LP

$$\begin{aligned} \text{minimize } & 1^T y = y_1 + \ldots + y_m \\ \text{subject to } & -y \leq x \leq y \\ & -1 \leq Ax - b \leq 1 \end{aligned}$$

with variables x, y.

e) Equivalent to the LP

$$\begin{aligned} \text{minimize } & 1^T y + t \\ \text{subject to } & -y \leq Ax - b \leq y \\ & -t_k \leq x_k \leq t_k, k = 1, \ldots, n. \end{aligned}$$

□

3.3 Power assignment in a wireless communication system.

We consider n transmitters with powers $p_1, \ldots, p_n \geq 0$, transmitting to n receivers. These powers are the optimization variables in the problem. We let $G \in R^{n \times n}$ denote the matrix of path gains from the transmitters to the receivers, $G_{ij} \geq 0$ is the path gain from transmitter j to receiver i.

The signal power at receiver i is then $S_i = G_{ii} p_i$, and the interference power at receiver i is

$$I_i = \sum_{k \neq i} G_{ik} p_k.$$

The *signal to interference plus noise ratio*, denoted SINR, at receiver i, is given by

$$SINR_i = \frac{S_i}{I_i + \sigma_i}$$

, where $\sigma_i > 0$ is the (self-) noise power in receiver i.

The objective in the problem is to maximize the minimum SINR ratio, over all receivers, *i.e.*,to maximize

$$\min_{i=1,\ldots,n} \frac{S_i}{I_i + \sigma_i}$$

There are a number of constraints on the powers that must be satisfied, in addition to the obvious one $pi \geq 0$. The first is a maximum allowable power for each transmitter, *i.e.*, $p_i \leq P_i^{\max}$, where $P_i^{\max} > 0$ is given.

In addition, the transmitters are partitioned into groups, with each group sharing the same power supply, so there is a total power constraint for each group of transmitter powers. More precisely, we have subsets $K_1, \ldots, K_m$ of $\{1, \ldots, n\}$ with $K_1 \cup \ldots \cup K_m = \{1, \ldots, n\}$ and $K_j \hat{K}_l = 0$ if $j \neq l$. For each group K_l, the total associated transmitter power cannot exceed $P_l^{gp} > 0$:

$$\sum_{k \in K_l} p_k \leq P_l^{gp}, l = 1, \ldots, m.$$

Finally, we have a limit $P_k^{rc} > 0$ on the total received power at each receiver:

$$\sum_{k=1}^{n} G_{ik} p_k \leq P_i^{rc}, i = 1, \ldots, n.$$

Formulate the SINR maximization problem as an optimization problem.

Proof.

$$\begin{aligned} \text{maximize } & \min_{i=1,\ldots,n} \frac{S_i}{I_i + \sigma_i} \\ \text{subject to } & 0 \leq p_i \leq P_i^{\max} \\ & \sum_{k \in K_l} p_k \leq P_l^{gp}, l = 1, \ldots, m, \\ & \sum_{k=1}^{n} G_{ik} p_k \leq P_i^{rc}, i = 1, \ldots, n. \end{aligned}$$

□

4 Duality and Optimal conditions

4.1 A simple example

Consider the optimization problem

$$\begin{array}{ll} \text{minimize} & x^2 + 1 \\ \text{subject to} & (x-2)(x-4) \leq 0, \end{array}$$

with variable $x \in \mathbb{R}$.

a) **Analysis of primal problem.** Give the feasible set, the optimal value, and the optimal solution.

b) **Lagrangian and dual function.** Plot the objective $x^2 + 1$ versus x. On the same plot, show the feasible set, the optimal point and value, and plot the Lagrangian $L(x, \lambda)$ versus x for a few positive values of λ. Verify the lower bound property

$$p^\star \geq \inf_x L(x, \lambda) \quad \text{for all } \lambda \geq 0.$$

Derive and sketch the Lagrange dual function $g(\lambda)$.

c) **Lagrange dual problem.** State the Lagrange dual problem associated with the given primal problem, and verify that it is a concave maximization problem.

Find the dual optimal value $d^\star$ and a dual optimal solution $\lambda^\star$.

Finally, check whether **strong duality** holds (i.e., whether $p^\star = d^\star$ and the optimal values are attained).

d) **Sensitivity analysis.** Let $p^\star(u)$ denote the optimal value of the perturbed problem

$$\begin{array}{ll} \text{min.} & x^2 + 1 \\ \text{s.t.} & (x-2)(x-4) \leq u, \end{array}$$

as a function of the parameter u.

Plot $p^\star(u)$. Verify that

$$\frac{dp^\star}{du}(0) = -\lambda^\star,$$

where $\lambda^\star$ is a dual optimal solution found in part (c).

4.2 QCQP, Slater's condition

$$\begin{array}{ll} \text{min.} & x_1^2 + x_2^2 \\ \text{s.t.} & (x_1-1)^2 + (x_2-1)^2 \leq 1 \\ & (x_1-1)^2 + (x_2+1)^2 \leq 1. \end{array}$$

with the variable $x \in R^2$.

a) Sketch the feasible set and level sets of the objective. Find the optimal point x^* and optimal value $p*$?

b) Give the KKT conditions. Verify with the results in question (a)?

c) Derive and solve the Lagrange dual problem. Does strong duality hold?

4.3 Equality constrained least-squares

$$\begin{aligned} &\text{minimize} \|Ax - b\|_2^2 \\ &\text{subject to } Gx = h. \end{aligned}$$

where $A \in R^{m \times n}$ with $\mathbf{rank} A = n$, and $G \in R^{p \times n}$ with $\mathbf{rank} G = p$. Give the KKT conditions, and derive expressions for the primal solution x^* and the dual solution ν^*.

4.4 QPQC

The problem

$$\begin{aligned} &\text{minimize} -3x_1^2 + x_2^2 + 2x_3^2 + 2(x_1 + x_2 + x_3) \\ &\text{subject to } x_1^2 + x_2^2 + x_3^2 = 1. \end{aligned}$$

is non-convex, however, the strong duality holds. Derive the KKT conditions. Find all solutions x, ν that satisfy the KKT conditions. Which pair corresponds to the optimum?

4.5 Quadratic program

Primal problem (assume $P \in S_{++}^n$)

$$\begin{aligned} &\text{minimize } x^T P x \\ &\text{subject to } Ax \leq b \end{aligned}$$

Answer Dual function:

$$g(\lambda) = \inf_x \left(x^T P x + \lambda^T (Ax - b) \right) = -\frac{1}{4} \lambda^T A P^{-1} A^T \lambda - b^T \lambda$$

Dual problem:

$$\begin{aligned} &\text{maximize } -\frac{1}{4} \lambda^T A P^{-1} A^T \lambda - b^T \lambda \\ &\text{subject to } \lambda \geq 0 \end{aligned}$$

from the Slater's condition: $p^\star = d^\star$ if the primal problem is feasible. In fact, $p^\star = d^\star$ always.

4.6 Quadratic problem with equality constraints

Derive the dual problem and KKT conditions at the optimal point.

$$\begin{aligned} &\text{minimize } x^T x \\ &\text{subject to } Ax = b \end{aligned}$$

4.7 Quadratic problem with equality constraints

Derive the dual problem and KKT conditions at the optimal point.

$$\begin{aligned} \text{minimize} \quad & \tfrac{1}{2} x^T P x + q^T x + r \\ \text{subject to} \quad & Ax = b, \end{aligned}$$

where $P \in \mathbb{S}_+^n$ (so the problem is convex).

4.8 Quadratic problem

Find the dual problem of the following problem:

$$\begin{aligned} \min\ & x_1^2 + x_2^2 - 2x_1 \\ \text{s.t. } & 2x_1 + x_2 \geq 5 \\ & x_2 \leq 7. \end{aligned} \tag{34}$$

Answer Primal problem:

$$\begin{aligned} \min\ & x_1^2 + x_2^2 - 2x_1 \\ \text{s.t. } & 2x_1 + x_2 \geq 5 \\ & x_2 \leq 7. \end{aligned} \tag{35}$$

Lagrangian function:

$$\begin{aligned} L(x_1, x_2, \lambda_1, \lambda_2) = & x_1^2 + x_2^2 - 2x_1 \\ & + (-2x_1 - x_2 + 5)\lambda_1 \\ & + (x_2 - 7)\lambda_2. \end{aligned}$$

Dual function:

$$\begin{aligned} g(\lambda_1, \lambda_2) &= \min_{x_1, x_2} L(x_1, x_2, \lambda_1, \lambda_2) \\ &= \min_{x_1} x_1^2 - 2(1 + \lambda_1)x_1 \\ &\quad + \min_{x_2} x_2^2 + (-\lambda_1 + \lambda_2)x_2 \\ &\quad + 5\lambda_1 - 7\lambda_2. \end{aligned}$$

Computation of the minimizers of the Lagrangian with respect to the primal variables $\operatorname{argmin}_{x_1,x_2} L$:

$$x_1 = 1 + \lambda_1$$
$$x_2 = \frac{1}{2}(\lambda_1 - \lambda_2).$$

Dual function:

$$g(\lambda_1, \lambda_2) = -(1+\lambda_1)^2 - \frac{1}{4}(\lambda_1 - \lambda_2)^2 + 5\lambda_1 - 7\lambda_2.$$

4.9 Minimizing negative entropy (Entropy maximization)

Find the dual problem of the following problem:

$$\begin{aligned} \min\ & f_0(x) = \sum_{i=1}^n x_i \log x_i \\ \text{s.t. } & Ax \leq b \\ & 1^T x = 1, \end{aligned}$$

where $A \in R^{r\times n}$.

Answer

$$\begin{aligned} \min\ & f_0(x) = \sum_{i=1}^n x_i \log x_i \\ \text{s.t. } & Ax \leq b \\ & 1^T x = 1, \end{aligned}$$

where $A \in \mathbf{R}^{r\times n}$.

Lagrangian:

$$\begin{aligned} L(x,\lambda,\nu) &= \sum_{i=1}^n x_i \log x_i + \lambda^T(Ax - b) + \nu(\sum_{i=1}^n x_i - 1) \\ &= \sum_{i=1}^n \Big(x_i \log x_i + (a_i^T\lambda + \nu)x_i\Big) - b^T\lambda - \nu. \end{aligned}$$

$$\nabla_{x_i} L(x,\lambda,\nu) = \log x_i + 1 + a_i^T\lambda + \nu = 0$$

when $x_i = a_i^T\lambda - \nu - 1$, substituting these x_i back into the Lagrangian yields the dual function

$$\begin{aligned} g(\lambda,\nu) &= -b^T\lambda - \nu - \sum_{i=1}^n e^{-a_i^T\lambda - \nu - 1} \\ &= -b^T\lambda - \nu - e^{-\nu-1}\sum_{i=1}^n e^{-a_i^T\lambda}, \end{aligned}$$

where a_i (or equivalently $a_{:,i}$) is the i-th column of the matrix A.

4.10 LP

Find the dual problem of the LP:

$$\min\ c^T x \tag{36}$$
$$\text{s.t. } Ax = B \tag{37}$$
$$x \geq 0. \tag{38}$$

Answer Consider the LP problem in general form

$$\min\ c^T x \tag{39}$$
$$\text{s.t. } Ax = B \tag{40}$$
$$x \geq 0. \tag{41}$$

Dual function is given by:

$$\begin{aligned} g(\lambda, \nu) &= \min_x L(x, \lambda, \nu) = -b^T\nu + \min_x (c + A^T\nu - \lambda)^T x \\ &= \begin{cases} -b^T\nu \ if\ c + A^T\nu - \lambda = 0 \\ -\infty, \text{otherwise} \end{cases} \end{aligned}$$

$-\infty$ is a trivial lower bound. Therefore, for the dual function $g(\lambda, \nu)$ to provide a genuine (finite) lower bound, we must impose the condition

$$c + A^T\nu - \lambda = 0.$$

The dual problem of the LP problem (39) is to find the best lower bound

$$\begin{aligned} &\max -b^T\nu \\ &\text{s.t. } A^T\nu - \lambda + c = 0 \\ &\quad \lambda \geq 0. \end{aligned}$$

equivalently,

$$\begin{aligned} &\max -b^T\nu \\ &\text{s.t. } A^T\nu + c \geq 0. \end{aligned}$$

4.11 LP

Find the dual problem of the LP:

$$\begin{aligned} &\min\ c^T x \\ &\text{s.t. } Ax \geq b \end{aligned}$$

Answer Consider the LP problem:

$$\begin{aligned} &\min\ c^T x \\ &\text{s.t. } Ax \geq b \end{aligned}$$

Dual function is given by:

$$\begin{aligned} g(\lambda) &= \min_x L(x,\lambda) = -b^T\lambda + \min_x (A^T\lambda + c)^T x \\ &= \begin{cases} -b^T\lambda \ if\ A^T + c\lambda = 0 \\ -\infty,\ if\ A^T + c\lambda \neq 0 \end{cases} \end{aligned}$$

The dual problem of the problem (**??**) is given by:

$$\begin{aligned} &\max -b^T\lambda \\ &\text{s.t. } A^T\lambda + c = 0 \\ &\lambda \geq 0. \end{aligned}$$

4.12 Quadratic constraint - quadratic programming (QCQP)

$$\begin{aligned} &\min\ x_1^2 + x_2^2 \\ &\text{s.t. } (x_1-1)^2 + (x_2-1)^2 \leq 1 \\ &\quad (x_1-1)^2 + (x_2+1)^2 \leq 1 \end{aligned}$$

a) Find the KKT conditions for the optimal point? Why the KKT conditions do not hold for this problem?

b) Find the dual problem?

4.13 simple Quadratic Problem

Consider the following problem:

$$\begin{aligned} &\min\ x^2 + 1 \\ &\text{s.t. } (x-2)(x-4) \leq 0, \end{aligned}$$

where $x \in R$.

a) Graphically show the feasible set, the optimal point and the minimum value of the problem?

b) Find the Lagrangian $L(x,\lambda)$ function, dual function $g(\lambda)$ and the dual problem? Verify that the dual problem is a concave maximization problem. Does the strong duality hold?

c) Plot $x^2 + 1$ vs. x, and $L(x,\lambda)$ vs. x on the same plot with several $\lambda \geq 0$? Verify the lower bound property $p^* \geq \min_x L(x,\lambda)$.

4.14 Simple LP problem

Find the dual problem of the following LP:

$$\begin{aligned} \min\ & x_1 + 2x_2 - x_3 \\ \text{s.t. } & x_1 + x_2 + x_3 = 1 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

4.15 LP

Find the dual problem of the following LP:

$$\begin{aligned} \min\ & c^T x \\ \text{s.t. } & Ax = b \\ & x \geq 0, \end{aligned}$$

where $x \in \mathbb{R}^n$.

4.16 LP

Find the dual problem of the following problem:

$$\begin{aligned} \min\ & y \\ \text{s.t. } & a_i^T x - b_i \leq 0, i = 1, \ldots, m \end{aligned}$$

4.17 Channel capacity

Find the dual problem of the following problem:

$$\begin{aligned} \min\ & -c^T x + \sum_{i=1}^{m} y_i \log(y_i) \\ \text{s.t. } & Px = y \\ & \mathbf{1}^T x = 1, \\ & x \geq 0 \end{aligned}$$

4.18 Least-square with linear equations

Consider the least-square with equality constraint:

$$\min\ x^\top x \tag{42}$$

$$\text{s.t. } Ax = b, \tag{43}$$

where $A \in \mathbb{R}^{m \times n}$ with rank$A = n$ and $G \in \mathbb{R}^{p \times n}$ with rank$A = m$. Write the KKT conditions and find the closed-form solution of the above problem from the KKT conditions.

4.19 Least-square with linear equations

Consider the least-square with equality constraint:

$$\min \|Ax - b\|_2^2 \tag{44}$$
$$\text{s.t. } Gx = h, \tag{45}$$

where $A \in \mathbb{R}^{m \times n}$ with rank$A = n$ and $G \in \mathbb{R}^{p \times n}$ with rank$G = p$. Write the KKT conditions and find the closed-form solution of the above problem from the KKT conditions.

4.20 Simple Quadratic

Consider the problem

$$\begin{aligned} \text{minimize} \quad & x_1^2 + x_2^2 \\ \text{subject to} \quad & x_1 + 2x_2 \geq 4. \end{aligned}$$

a. Give the feasible set, optimal value, and optimal solution.

b. Derive the dual problem and solve the dual problem. Does the strong duality hold?

c. Verify the primal and dual optimal solutions satisfying KKT conditions.

d. Write the **dual-based gradient ascent method** to solve the problem.

4.21 LP

Find the dual of the Linear Programming:

$$\begin{aligned} \text{minimize} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} = \mathbf{b}, \\ & \mathbf{x} \geq \mathbf{0}, \end{aligned}$$

where $x \in \mathbf{R}^n$, $A \in \mathbf{R}^{m \times n}$, and $b \in \mathbf{R}^m$.

4.22 Water-filling

Consider the problem

$$\begin{aligned} x \quad & -\sum_{i=1}^{n} \log(\alpha_i + x_i) \\ & x \geq 0, \qquad \mathbf{1}^T x = 1, \end{aligned}$$

where $\alpha_i > 0$ for all i. This problem arises in information theory when allocating a fixed total transmit power (of one) among n parallel communication channels. The variable x_i represents the power allocated to the i-th channel, and $\log(\alpha_i + x_i)$ gives the capacity (or communication rate) of that channel. Thus the objective is to *maximize* the total communication rate.

Answer We introduce Lagrange multipliers $\lambda^\star \in \mathbb{R}^n$ for the inequality constraints $x \geq 0$, and a scalar multiplier $\nu^\star \in \mathbb{R}$ for the equality constraint $\mathbf{1}^T x = 1$. The KKT conditions are

$$\begin{aligned} x^\star \geq 0, \qquad \mathbf{1}^T x^\star = 1, \qquad \lambda^\star \geq 0, \\ \lambda_i^\star x_i^\star = 0, \quad i = 1, \ldots, n, \\ -\frac{1}{\alpha_i + x_i^\star} + \lambda_i^\star + \nu^\star = 0, \quad i = 1, \ldots, n. \end{aligned}$$

We can solve these equations directly to find $x^\star$, $\lambda^\star$, and $\nu^\star$. Observe that $\lambda^\star$ only appears in the complementarity conditions and acts as a slack variable in the stationarity equation, so it can be eliminated. This leaves

$$x^\star \geq 0, \quad \mathbf{1}^T x^\star = 1, \quad \lambda_i^\star\big(\nu^\star - 1/(\alpha_i + x_i^\star)\big) = 0, \quad i = 1, \ldots, n,$$

or equivalently

$$\nu^\star \geq \frac{1}{\alpha_i + x_i^\star}, \quad i = 1, \ldots, n.$$

If $\nu^\star < 1/\alpha_i$, this last condition can only hold if $x_i^\star > 0$, which by the third condition forces $\lambda_i^\star = 0$ and thus $\nu^\star = 1/(\alpha_i + x_i^\star)$. Solving for $x_i^\star$, we conclude that $x_i^\star = 1/\nu^\star - \alpha_i$ if $\nu^\star < 1/\alpha_i$. If $\nu^\star \geq 1/\alpha_i$, then $x_i^\star > 0$ is impossible (because it would imply $\nu^\star \geq 1/(\alpha_i + x_i^\star) > 1/\alpha_i$), violating complementary slackness. Therefore,

$$x_i^\star = \begin{cases} 1/\nu^\star - \alpha_i & \nu^\star < 1/\alpha_i, \\ 0 & \nu^\star \geq 1/\alpha_i. \end{cases}$$

or, more compactly,

$$x_i^\star = \max\big(0,\ 1/\nu^\star - \alpha_i\big).$$

Substituting this expression for $x_i^\star$ into the power-balance constraint $\mathbf{1}^T x^\star = 1$, we obtain

$$\sum_{i=1}^n \max\big(0,\ 1/\nu^\star - \alpha_i\big) = 1.$$

The left-hand side is a piecewise-linear increasing function of $1/\nu^\star$, with breakpoints at the values α_i, so the equation has a *unique* solution that is readily determined.

This solution method is known as the **water-filling algorithm** for the following reason. We think of α_i as the ground level above patch i, and we then flood the region with water to a depth $1/\nu^\star$, as illustrated in Figure 5.7. The total amount of water used is $\sum_i \max(0, 1/\nu^\star - \alpha_i)$. We then increase the flood level until we have used a total amount of water equal to one. The depth of water above patch i is then the optimal value $x_i^\star$.

5 First-Order Algorithms

5.1 LASSO

Consider linear regression with ℓ_1-norm regularization:

$$\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \lambda \|\mathbf{w}\|_1,$$

where $\lambda > 0$ is a constant and $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$.

a) Write the **sub-gradient** method to solve the above problem.

b) Write the **proximal gradient descent** method to solve the above optimization problem.

c) Write the **proximal Stochastic gradient descent** method to solve the above optimization problem.

5.2 logistic regression with ℓ_1-norm regularization

Given logistic regression with ℓ_1-norm regularization

$$\text{min. } \frac{1}{m} \sum_{i=1}^{m} \big(- y_i \mathbf{x}_i^T \mathbf{w} + \log(1 + e^{\mathbf{x}_i^T \mathbf{w}} + \lambda \|\mathbf{w}\|_1 \big),$$

where $\lambda > 0$.

a) Derive the Sub-gradient Descent Algorithm to solve the given optimization problem with a fixed regularization parameter λ. Clearly state the sub-gradient update rule for the weight vector $\mathbf{w}$.

b) Derive the Proximal Gradient Method (or FISTA) to solve the optimization problem. Specifically, derive the update rule for $\mathbf{w}$, clearly defining the proximal operator (or prox function) used in the derivation.

c) Derive the Proximal Stochastic Gradient Method (Prox-SGD) to solve the optimization problem. Use a random mini-batch of size B in the derivation and state the final update rule for the weight vector $\mathbf{w}$.

5.3 Quadratic problem with equality constraints

Consider the following problem

$$\begin{aligned} &\text{minimize } \|Ax - b\|_2^2 \\ &\text{s.t. } Gx <= h \end{aligned}$$

where f is convex and differentiable, and g is convex (possibly non-smooth). Write the **dual-based gradient ascent method** to solve the problem.

6 Second-order Algorithms

6.1 Simple problem, Newton method regression

Consider the unconstrained problem

$$\min \frac{10x_1^2 + x_2^2}{2} + 5\log(1 + e^{-x_1 - x_2}).$$

a) Describe **Newton's method** to solve the above problem (including the update rule, how to choose the step size, and stopping criterion).

b) Compare and contrast **first-order** and **second-order** methods (e.g., gradient descent vs. Newton's method) in terms of convergence rate, per-iteration cost, robustness, etc.

6.2 The pure Newton method

Newton's method with fixed step size $t = 1$ can diverge if the initial point is not close to x^*?. In this problem we consider two examples.

1. f(x) = $\log(e^x + e^x)$ has a unique minimizer $x^\star = 0$. Run Newton's method with fixed step size $t = 1$, starting at $x^{(0)} = 1$ and at $x^{(0)} = 1.1$.
2. $f(x) = \log x + x$ has a unique minimizer $x^\star = 1$. Run Newton's method with fixed step size $t = 1$, starting at $x^{(0)} = 3$.

6.3 Log-barrier method

Consider the following problem

$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) \\ \text{subject to} & A\mathbf{x} \leq \mathbf{b} \end{array}$$

where $\mathbf{x} \in \mathbb{R}^n$, $A \in \mathbb{R}^{p \times n}$ and $\mathbf{b} \in \mathbb{R}^p$. $f(\mathbf{x})$ is a convex and smooth function.

Give the log-barrier method solving the above problem.

Solution: The Log-Barrier (Interior-Point) Method

The problem has p linear inequality constraints, $a_i^T \mathbf{x} \leq b_i$, where a_i^T is the i-th row of A. We define the constraints $f_i(\mathbf{x}) \leq 0$:

$$f_i(\mathbf{x}) = a_i^T \mathbf{x} - b_i \leq 0, \quad \text{for } i = 1, \ldots, p$$

1. The Logarithmic Barrier Function ($\phi(\mathbf{x})$)

The logarithmic barrier function is applied to enforce the interior of the feasible set:

$$\phi(\mathbf{x}) = -\sum_{i=1}^{p} \log(-f_i(\mathbf{x})) = -\sum_{i=1}^{p} \log(b_i - a_i^T \mathbf{x})$$

This function is convex and smooth on the interior of the feasible set.

2. The Approximating Unconstrained Objective ($\Phi_t(\mathbf{x})$)

For a strictly positive barrier parameter $t > 0$, the constrained problem is approximated by the unconstrained minimization of $\Phi_t(\mathbf{x})$:

$$\Phi_t(\mathbf{x}) = tf(\mathbf{x}) + \phi(\mathbf{x}) = tf(\mathbf{x}) - \sum_{i=1}^{p} \log(b_i - a_i^T \mathbf{x})$$

The solution to this problem, $\mathbf{x}^*(t)$, is the **central point**.

3. The Log-Barrier Algorithm

The algorithm solves the original problem by repeatedly minimizing $\Phi_t(\mathbf{x})$ for an increasing sequence of t.

Input: Strictly feasible starting point $\mathbf{x}^{(0)}$, initial $t^{(0)} > 0$, multiplier $\mu > 1$, tolerance $\epsilon > 0$.

Initialization: Set $\mathbf{x} = \mathbf{x}^{(0)}$ and $t = t^{(0)}$.

Repeat (Outer Iteration):

1. **Centering Step (Inner Loop):** Compute the central point $\mathbf{x}^*(t)$ by minimizing $\Phi_t(\mathbf{x})$, using the current $\mathbf{x}$ as a starting point for Newton's method.
$$\mathbf{x} := \arg\min_{\mathbf{z}} \left(tf(\mathbf{z}) - \sum_{i=1}^{p} \log(b_i - a_i^T \mathbf{z}) \right)$$
2. **Stopping Check:** If the duality gap $\frac{p}{t} < \epsilon$, then terminate. The current $\mathbf{x}$ is the approximate solution.
3. **Increase t:** Update the barrier parameter:
$$t := \mu t$$

References

[1] Boyd, Stephen, Stephen P. Boyd, and Lieven Vandenberghe. Convex optimization. Cambridge university press, 2004.

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